European Network of
Transmission System Operators
For Electricity

Operational Manual

Energy Communication Platform
Version 4.16.0

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Energy Communication Platform, version 4.16.0

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2. Executive Summary

This document provides the operational manual for all ECP components.

This document is targeted at Administrators and Business Users.

3. Revision History

Version	Date	Author	Description
4.10.1	16.01.2023	Drahoslav Stejskal	Document Creation
4.11.0	16.05.2023	Jiří Kakrda	Version update
4.12.0	27.10.2023	Jiří Kakrda	Version update
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4.15.0	21.10.2024	Jiří Kakrda	Corrupted CD xml behaviour description
4.16.0	06.05.2025	Jiří Kakrda	Version update

4. ECP Components

Energy Communication Platform is internally composed of four main components – Endpoint, Internal Broker, Component Directory, and Broker.

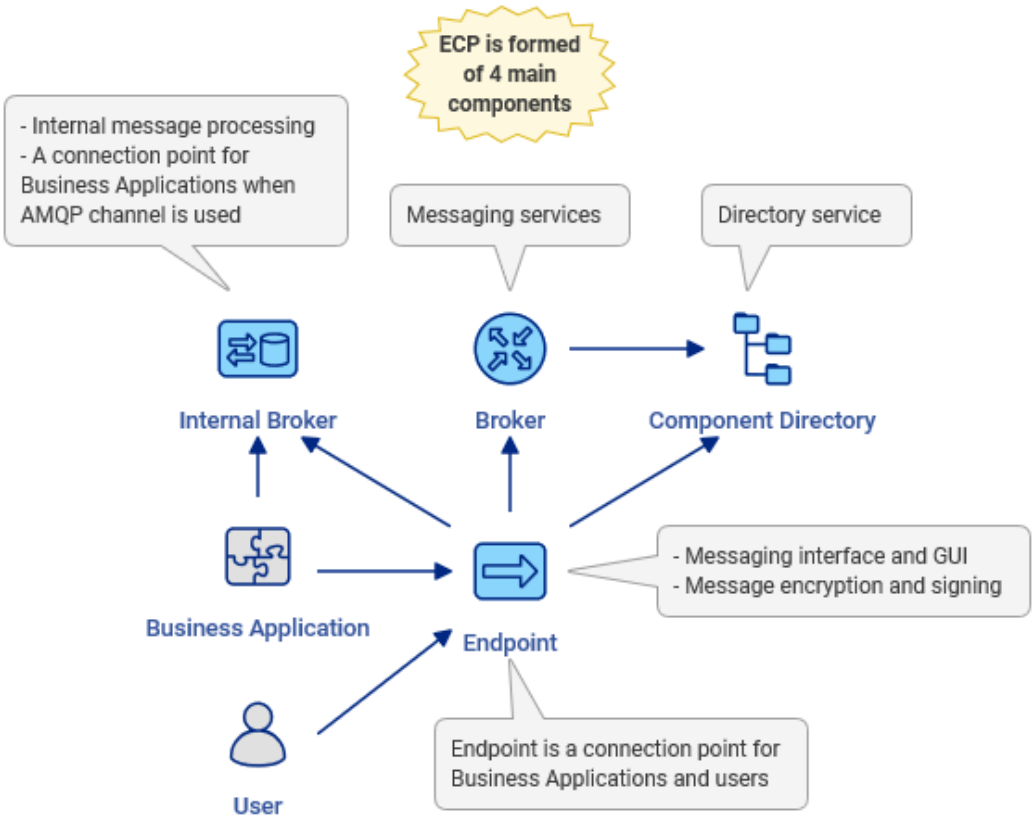

Figure 1: Main Components

4.1. Endpoint

From the users' point of view, the crucial component is the Endpoint, which provides the user interface for sending and receiving messages and the API for integration with business applications.

The ECP Endpoint is connected to Internal Broker, which serves for internal messaging service. Connection from ECP Endpoint to Internal Broker is realized via AMQPS protocol which usually uses default port 5672.

The ECP Endpoint is also connected to ECP Broker, which serves for external messaging service. ECP Broker for messaging is defined by message path, which is configured on ECP Endpoint recipient side. Connection from ECP Endpoint to ECP Broker is realized via AMQPS protocol which usually uses default port 5671.

To obtain certificates and information about other ECP components in ECP network, ECP Endpoint needs to be connected to ECP Component Directory (CD). Connection from ECP Endpoint to CD is realized via HTTPS protocol with default port 8443. Each component must be first registered in a Component Directory, referred as its home CD, to be able to exchange messages in the ECP network. Connection to CD is not mandatory for messaging, until the TTL record expires. Default TTL record is 28 days for ECP CD.

4.2. Internal Broker

The Internal Broker provides messaging services for the internal message processing in ECP Endpoint. Endpoint must be always connected to Internal Broker to be able to process messages. Multiple Endpoints can be connected to one Internal Broker instance. Business Application can also send and receive messages from Internal Broker in case the AMQP integration channel is used.

4.3. Component Directory

Component Directory is a standalone component containing key information about all the components registered into the ECP network, which are Endpoints, Brokers, and other Component Directories.

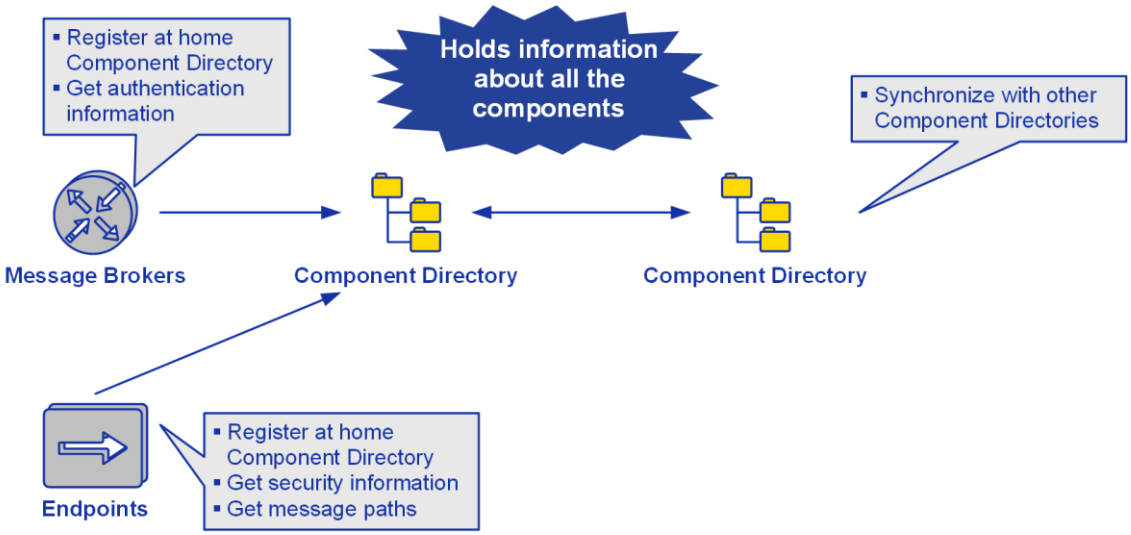

Figure 2: ECP High-level Architecture - Component Directory

The key responsibility and features of the Component Directory are:

- Provide registration interface to the Endpoints and the Brokers, which use it to request a new registration within their home Component Directory or renew certificates for an already registered component.
- Issue the certificates used by the ECP components.
- Maintain security information (signing, encryption, and authentication certificates) about all the Endpoints in the network.
- Maintain and provide the message paths established between the Endpoints.
- Mutually synchronize the Directory content with other Component Directories to maintain consistent information throughout the ECP network.
- Provide the synchronization interface which is used by other ECP components.

Note: Short unavailability of Component Directory does not affect ECP messaging. Connection to CD is not mandatory for messaging, until the TTL record expires. Default TTL record is 28 days. To get further information about configuration of TTL for ECP components, please refer to the *ECP Administration Guide, chapter How to configure components expiration*.

4.4. Message Broker

The Message Broker (or simply a Broker) is a router component, used by the Endpoints, that acts as an optional intermediate component in the message route path. It ensures a reliable and secure communication with the Endpoints using SSL protocol and it authenticates the connecting Endpoints using their authentication certificate.

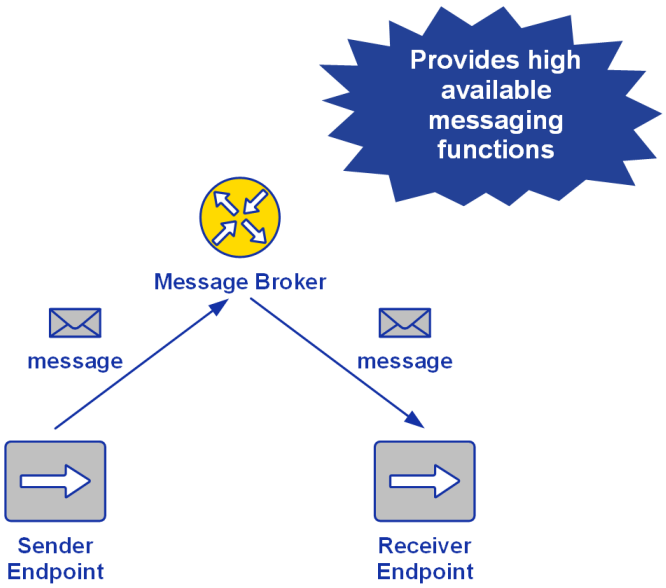

Figure 3: Message Broker

Note: AMQPS connection from ECP Endpoints to ECP Broker is crucial for ECP messaging.

5. Starting and stopping the ECP services

ECP applications can be managed via services or using a command line. For management of services on Linux platform, the command systemctl is used. On Windows platform, ECP services are available via Windows Services Manager - services.msc.

5.1. Starting and stopping ECP Endpoint on Linux

- When ECP Endpoint is registered as a Linux service, the following commands are available:
  - To start ECP, use command "*systemctl start ecp-endpoint.service*"
  - To stop ECP, use command "*systemctl stop ecp-endpoint.service*"
  - To restart ECP, use command "*systemctl restart ecp-endpoint.service*"
  - To check the service status, use command "*systemctl status ecp-endpoint.service*"
- If not installed as a Linux service:
  - Start application using bash script "*<install_path>/tomcat/bin/startup.sh*"
  - Stop application using bash script "*<install_path>/tomcat/bin/shutdown.sh*"

5.2. Starting and stopping ECP Endpoint on Windows

5.2.1. <u>Using GUI</u>

- If installed as a service:
  - Start > Run > "*services.msc*"
  - Find service with name like "*ecp-endpoint*" and description "*ecp-endpoint*"
  - Start / stop service by buttons on the top left
  - Check the service status in the *Status* column

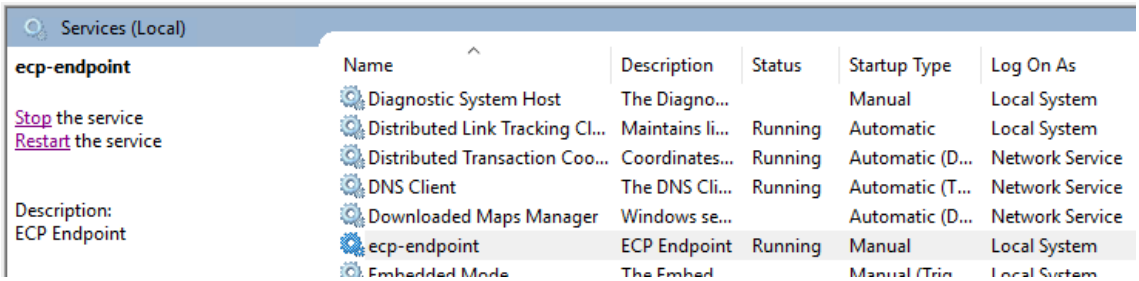

Figure 4: Endpoint Service

- If not installed as a service:
  - Start / stop application by using
    - *<install_path>\tomcat\bin\startup.bat*
    - *<install_path>\tomcat\bin\shutdown.bat*

5.2.2. Using Command Line

- If installed as a service:
  - Start service using:
    - *SC start ecp-endpoint*
  - Stop service using:
    - *SC stop ecp-endpoint*
- If not installed as a service:
  - Start application using:
    - *<install_path>\tomcat\bin\startup.bat*
  - Stop application using:
    - *<install_path>\tomcat\bin\shutdown.bat*

Note: For more information about Endpoint services management possibilities, please refer to the *ECP Installation Guide, chapters Starting and stopping ECP Endpoint.*

5.3. Starting and stopping Internal Broker on Linux

- When Internal Broker is registered as a Linux service, the following commands are available:
  - To start Internal Broker, use command "*systemctl start* eccosp-artemis.*service*"
  - To stop Internal Broker, use command "*systemctl stop* eccosp-artemis.*service*"
  - To restart Internal Broker, use command "*systemctl restart* eccosp-artemis.*service*"
  - To check the service status, use command "*systemctl status* eccosp-artemis.*service*"
- If not installed as a Linux service:
  - Start application using bash script "/opt/eccosp-artemis/bin/artemis run"
  - Stop application using bash script "/opt/eccosp-artemis/bin/artemis stop"

5.4. Starting and stopping Internal Broker on Windows

5.4.1. Using GUI

- If installed as a service:
  - Start > Run > "*services.msc*"
  - Find service with name like *"eccosp-artemis"*
  - Start / stop service by buttons on the top left
  - Check the service status in the *Status* column

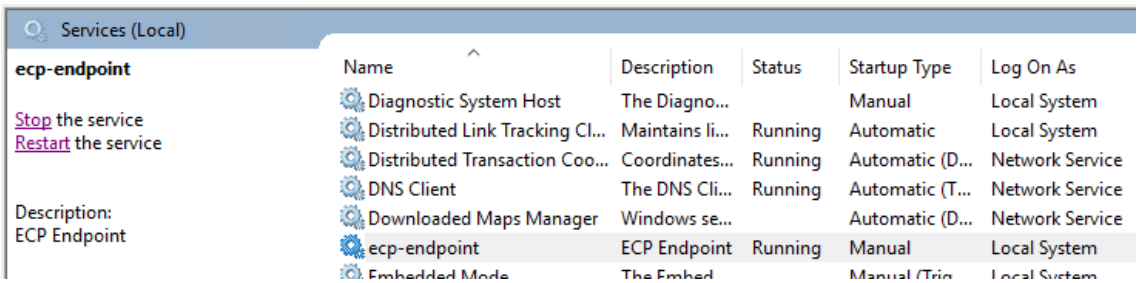

Figure 5: Internal Broker Service

- If not installed as a service:

- Start / stop application by using
  - <install_path>\broker\bin\artemis.cmd run
  - <install_path>\broker\bin\artemis.cmd stop

5.4.2. Using Command Line

- If installed as a service:
  - Start service using:
    - *SC start artemis-broker-0.0.0.0*
  - Stop service using:
    - *SC stop artemis-broker-0.0.0.0*
- If not installed as a service:
  - Start application using:
    - *<install_path>\broker\bin\artemis.cmd run*
  - Stop application using:
    - *<install_path>\broker\bin\artemis.cmd stop*

Note: For more information about Internal Broker services management possibilities, please refer to the *ECP Installation Guide, chapters Starting and stopping Internal Broker.*

5.5. Starting and stopping ECP Component Directory

- When ECP Component Directory is registered as a Linux service, the following commands are available:
  - To start ECP, use command "*systemctl start ecp-directory.service*"
  - To stop ECP, use command "*systemctl stop ecp-directory.service*"
  - To restart ECP, use command "*systemctl restart ecp-directory.service*"
  - To check the service status, use command "*systemctl status ecp-directory.service*"
- If not installed as a Linux service:
  - Start application using bash script "*<install_path>/tomcat/bin/startup.sh*"
  - Stop application using bash script "*<install_path>/tomcat/bin/shutdown.sh*"

Note: For more information about Component Directory services management possibilities, please refer to the *ECP Installation Guide, chapters Starting and stopping ECP Component Directory.*

5.6. Starting and stopping ECP Broker

- To start ECP, use command "*systemctl start ecp-broker.service*"
- To stop ECP, use command "*systemctl stop ecp-broker.service*"
- To restart ECP, use command "*systemctl restart ecp-broker.service*"
- To check the service status, use command "*systemctl status ecp-broker.service*"

Note: For more information about Broker services management possibilities, please refer to the *ECP Installation Guide, chapters Starting and stopping ECP Broker.*

6. Health Check

This chapter describes basic health checks for all ECP components. Please note, that the manual health check is just a supplement to a complex monitoring system. Please, see chapter below the table to gather information about health check instructions.

Note: For more information about ECP Monitoring possibilities, please refer to the *ECP Administration Guide, chapter Monitoring.*

6.1. Endpoint

ID	Condition	Action point if condition is NOT fulfilled
1	☐ ECP Endpoint GUI is reachable.	☐ Check ecp-endpoint service status. Start ECP Endpoint service if not running. ☐ Analyze Application and Catalina logs to find a root cause. ☐ Check whether and on which interface the ECP Endpoint is listening – for instance, using netstat command. ☐ If service is running and no issue is recorded in log file, check network rules if connection to ECP Endpoint GUI is allowed.
2	☐ Readiness Actuator status is 'READY'.	☐ Check Readiness Actuator status detail. ☐ Analyze Application log to find a root cause.
3	☐ Connectivity test to Component Directory returns 'CONNECTED'.	☐ Analyze Application log to find a root cause of connectivity issue. ☐ If possible, check connectivity to the Component Directory from other ECP Endpoints to rule out a local issue. ☐ Try to reach the Component Directory URL via web browser or some network tool – e.g.: curl, telnet, wget, etc. ☐ Check network rules if HTTPS connection From ECP Endpoint to Component Directory is allowed. ☐ Report possible issue on central component to the project owner or contact Component Directory administrator if no local issue is identified.

4	☐ Messaging connectivity test to other ECP Endpoint(s) returns 'CONNECTED'.	☐ Try messaging connectivity to several different ECP Endpoints. Make sure that the used message type is in align with message path configuration on recipient side. ☐ Analyse Application log to find a root cause of connectivity issue. ☐ Check connectivity to Messaging Broker using a network tool e.g.: curl, telnet. ☐ Check network rules if AMQPS connection From ECP Endpoint to Messaging Broker is allowed. ☐ Report possible issue on central component to the project owner or contact Component Directory administrator if no local issue is identified.
5	☐ Certificates expiration is not sooner than 30 days, certificates are not expired.	☐ Make sure that Automatic Renewal feature for ECP Endpoint is Enabled. ☐ Renew certificates manually.
6	☐ Logs do not contain any unexpected ERRORs or exceptions.	☐ Analyse Application and Catalina logs to find a root cause.

Note: In case that the ECP Endpoint is not working and action points in the table above did not help to resolve the issue, please contact your local ECP support.

6.1.1. Endpoint Health Check Instructions

6.1.1.1. ECP Endpoint GUI is reachable

- ECP Endpoint GUI can be accessed via:
  - https://<endpoint-ip>:8443/ECP_MODULE

6.1.1.2. Check ecp-endpoint service status

- Please refer to chapters *Starting and stopping the ECP Endpoint on Linux*, and *Starting and stopping the ECP Endpoint on Windows*.

6.1.1.3. Check Application and Catalina logs

- In case of any issue, it is recommended to check the logs. Logs are located in */var/log/ecp-endpoint/* folder (For Windows: *<install_path>\tomcat\logs*).
- ECP Endpoint uses application, audit and Tomcat catalina logs. Tomcat Catalina log contains mostly events connected with starting and stopping of the application. For example, when the application is not able to start due to memory issue, this information will be logged there.
- Application log contains all events related to application behavior such as message processing, background jobs and others.
- In case that audit logging is enabled, the most of log events is available in audit log file. Audit log contains various events regarding sending or receiving messages, validations, user activities on user interface and much more. List of audit log events can be found in the audit log excel, which is part of the ECCo SP Documentation Delivery.

6.1.1.4. Readiness Actuator status is 'READY'

- ECP Endpoint provides the Readiness actuator. The actuator is checking ECP Endpoint's readiness every 5 seconds. The following conditions are checked:
  - Endpoint is in ACTIVE state
  - Authentication certificate is not expired
  - Endpoint is not reloading its configuration
  - Endpoint is connected to its Internal Broker
  - Endpoint is synchronized with Component Directory
  - There is at least one valid message path

- In case of issue, for example ECP Endpoint is not able to connect to its internal broker, the actuator will respond with status NOT_READY and description of the problem.
- Actuator's response can be checked by *<ecp-address>/ECP_MODULE/actuator/readiness* request or on GUI, on the dashboard page. Please note, that the readiness endpoint must be enabled in the configuration.
- To get further information about readiness actuator, please refer to the *ECP Administration Guide, chapter Readiness Endpoint*.

6.1.1.5. Connectivity test to Component Directory

- To test the connectivity between ECP Endpoint and ECP Component Directory, go to the Settings page, section Component Directory and click the Check Connectivity button. When connectivity test is done, the status will be displayed right next to the button.
- To get further information about connectivity test, please refer to the *ECP Functional Specification, chapter UC Connectivity Test*.

6.1.1.6. Messaging connectivity test to other ECP Endpoint(s)

- To test the messaging connectivity, go to the settings page, section Messaging connectivity, and click the Check Connectivity button. This button will redirect you to the Connectivity Check page. Choose Receiver's ECP Endpoint code from the dropdown menu, fill the message type you want to test and click the Check button. When connectivity check is done, the status will be displayed right above the Check button.
- To get further information about connectivity test, please refer to the *ECP Functional Specification, chapter UC Connectivity Test*.

6.1.1.7. Certificates expiration check, renewal

- Information about ECP certificates is available via Certificates tile on Dashboard page.
- The Certificates tile displays number of days to the closest expiration with expiration date and certificate type. This tile will change its colour to red when renewal threshold is reached.
- Detail information about certificates is available in '*Settings'* tab in section '*Certificates'*.
- In the Certificates section, Automatic Renewal can be enabled by clicking on "*Enabled*" button.

- In the Certificates section, the certificates can be manually renewed by clicking on button "*Renew Manually*".
- To get further information about certificates renewal, please refer to the *ECP Administration Guide, chapter Certificates renewal*.

6.1.2. Corrupted CD content XML stored in DB

When the ECP Endpoint is started and a corrupted ECP Component Directory XML content is stored in the database, the ECP Endpoint will start in the anonymous mode. ECP Endpoint will refresh the ECP Component Directory XML content after the next successful synchronization with the ECP Component Directory. Registration Status Checker background job can fail and log an exception, while the ECP Component Directory XML content was not renewed yet.

6.2. Internal Broker

ID	Condition	Action point if condition is NOT fulfilled
1	☐ Artemis Console GUI is reachable.	☐ Check eccosp-artemis service status. Start Internal Broker service if not running. ☐ Analyze Artemis log to find a root cause. ☐ Check whether and on which interface the Artemis is listening – for instance, using netstat command. ☐ Check on which host is configured Artemis console. ☐ If service is running and no issue is recorded in log file, check network rules if connection to Artemis console is allowed.
2	☐ Check that queues in Artemis console (except DLQ) have a consumer. Make sure that Message Count is 0 or decreasing.	☐ Restart eccosp-artemis service. ☐ If a consumer is still disconnected after service restart, validate network rules. Incoming connection to port 5672 needs to be allowed. ☐ Contact owner of ECP Endpoint to restart ecp-endpoint service and check their logs for a root cause analysis.
3	☐ Logs do not contain any unexpected ERRORs or exceptions.	☐ Analyse Artemis log to find a root cause.

6.2.1. Internal Broker Health Check Instructions

6.2.1.1. Artemis Console GUI is reachable

- Artemis Management Console GUI can be accessed via:
  - http://<internal-broker-ip>:8161/
- To get further information about Artemis GUI, please refer to the *ECP Installation Guide, chapter Artemis GUI*.

6.2.1.2. Check eccosp-artemis service status

- Please refer to the chapters *Starting and stopping Internal Broker*.

6.2.1.3. Check Artemis logs

- In case of any issue, it is recommended to check the logs. Logs are located in */opt/eccosp-artemis/log* folder *(<install_path>\broker\log* in Windows).
- Application log (Artemis) contains all events related to application behavior such as incoming connections.

6.2.1.4. Artemis console – check of connections from other ECP Endpoints

- Connect to Artemis console on URL *http://<ecp-broker-ip>:8161/*, from the home screen click on the Queues item from the upper menu.
- Check that queues in Artemis console (except DLQ) have a consumer (Consumer Count). Make sure that Message Count is 0 or decreasing.
- If a consumer is disconnected, restart eccosp-artemis service.
- If a consumer is still disconnected after service restart, validate network rules. Incoming connection to port 5672 needs to be allowed.
- Contact owner of ECP Endpoint to restart their ecp-endpoint service and check their logs for a root cause analysis.

6.3. Component Directory

ID	Condition	Action point if condition is NOT fulfilled
1	☐ Component Directory GUI is reachable.	☐ Check ecp-directory service status. Start ECP Directory service if not running. ☐ Analyze Application and Catalina logs to find a root cause. ☐ Check whether and on which interface the Component Directory is listening – for instance, using netstat command. ☐ If service is running and no issue is recorded in log file, check network rules if connection to Component Directory GUI is allowed.
2	☐ Readiness Actuator status is 'READY'.	☐ Check Readiness Actuator status detail. ☐ Analyze Application log to find a root cause.
3	☐ ECP Endpoint(s) last synchronisation with CD is up to date – not older than 5 minutes.	☐ Check if the issue is valid for all localy registered ECP Endpoints or if it could be a local issue on ECP Endpoint(s) side. ☐ Analyze Application log to find a root cause of connectivity issue. ☐ Check network rules if HTTPS connection to Component Directory is allowed.
4	☐ Synchronization with other Component Directories returns 'OK'.	☐ Analyze Application log to find a root cause of connectivity issue. ☐ If possible, check connectivity to the remote Component Directory from other ECP Endpoints to rule out a local issue. ☐ Try to reach the remote Component Directory URL via web browser or some network tool – e.g.: curl, telnet, wget, etc.

		☐ Check network rules if HTTPS connection from local Component Directory to a remote Component Directory is allowed. ☐ Report possible issue on central component to the Component Directory owner/administrator if no local issue is identified.
5	☐ Certificates expiration is not sooner than 30 days, certificates are not expired.	☐ Make sure that Automatic Renewal feature for Component Directory is Enabled. ☐ Renew certificates manually.
6	☐ Logs do not contain any unexpected ERRORs or exceptions.	☐ Analyze Application and Catalina logs to find a root cause.

Note: In case that the ECP Component Directory is not working and action points in the table above did not help to resolve the issue, please contact your local ECP support.

6.3.1. Component Directory Health Check Instructions

6.3.1.1. Component Directory GUI is reachable

- ECP Component Directory GUI can be accessed via:
  - https://<component-directory-ip>:8443/ECP_MODULE

6.3.1.2. Check ecp-directory service status

- Please refer to chapters *Starting and stopping the ECP Component Directory on Linux* and *Starting and stopping the ECP Component Directory on Windows*.

6.3.1.3. Check Application and Catalina logs

- In case of any issue, it is recommended to check the logs. Logs are located in */var/log/ecp-directory/* folder (For Windows: *<install_path>\tomcat\logs*\).
- Component Directory uses application, audit, access and Tomcat catalina logs. Tomcat Catalina log contains mostly events connected with starting and stopping of the application. For example, when the application is not able to start due to memory issue, this information will be logged here.
- Application log contains all events related to application behaviour such as message processing, background jobs and others.
- Audit log contains various events regarding synchronisation or managing ECP certificates, user activities on user interface and much more. List of audit log events can be found in the audit log excel, which is part of the ECCo SP Documentation Delivery.
- Access log contains information about all interactions with Component Directory GUI and API, all synchronization requests are recorded.

6.3.1.4. Readiness Actuator status is 'READY'

- ECP Component Directory provides the Readiness actuator. The actuator is checking ECP Directory's readiness every 5 seconds. The following conditions are checked:
  - Component Directory is registered (authentication certificate exists in the database)
  - Authentication certificate is not expired
  - Component Directory is not reloading its configuration
- In case of issue, for example ECP Component Directory is expired, the actuator will respond with status NOT_READY and description of the problem.

- Actuator's response can be checked by *<ecp-address>/ECP_MODULE/actuator/readiness* request or on GUI, on the dashboard page. Please note, that the readiness endpoint must be enabled in the configuration.
- To get further information about readiness actuator, please refer to the *ECP Administration Guide, chapter Readiness Endpoint*.

6.3.1.5. Certificates expiration check, renewal

- Information about Component Directory certificates is available via Certificates tile on Dashboard page.
- The Certificates tile displays number of days to the closest expiration with expiration date and certificate type. This tile will change its colour to red when renewal threshold is reached.
- Detail information about certificates is available in '*Settings'* tab in section '*Certificates'*.
- Make sure that automatic renewal is enabled on Component Directory. To get further information about certificates renewal, please refer to the *ECP Administration Guide, chapter Certificates renewal*.
- In the Certificates section, a new certificate can be imported by click on button "*Import Local Certificate*".

6.4. Message Broker

ID	Condition	Action point if condition is NOT fulfilled
1	☐ Artemis Console GUI is reachable.	☐ Check ecp-broker service status. Start ECP Broker service if not running. ☐ Analyze Artemis log to find a root cause. ☐ Check whether and on which interface the Artemis is listening – for instance, using netstat command. ☐ Check on which host is configured Artemis console. ☐ If service is running and no issue is recorded in log file, check network rules if connection to Artemis console is allowed.
2	☐ Check that queues in Artemis console (except DLQ) have a consumer. Make sure that Message Count is 0 or decreasing.	☐ If lot of clients is disconnected, restart ecp-broker service. ☐ If majority of clients is still disconnected after service restart, validate network rules. Incoming connection to port 5671 needs to be allowed. ☐ If just several clients are disconnected or not consuming messages from queue, contact owner of ECP Endpoint to restart ecp-endpoint service and check their logs for a root cause analysis.
3	☐ Folder for ECP Component Directory content storage contains a file not older than 5 minutes (default sync interval).	☐ Analyze Artemis log to find a root cause of connectivity issue. ☐ If possible, check connectivity to the Component Directory from other ECP Endpoints to rule out a local issue. ☐ Try to reach the Component Directory URL via web browser or some network tool – e.g.: curl, telnet, wget, etc.

		☐ Check network rules if HTTPS connection From Messager Broker to Component Directory is allowed. ☐ Report possible issue on central component to the project owner or contact Component Directory administrator if no local issue is identified.
4	☐ Certificates expiration is not sooner than 30 days, certificates are not expired.	☐ Renew certificates manually if certificates are not expired yet. ☐ Send a new registration request to Component Directory if the certificates are already expired.
5	☐ Logs do not contain any unexpected ERRORs or exceptions.	☐ Analyse Artemis log to find a root cause.

Note: In case that the ECP Broker is not working and action points in the table above did not help to resolve the issue, please contact your local ECP support.

6.4.1. Message Broker Health Check Instructions

6.4.1.1. Artemis Console GUI is reachable

- Artemis Management Console GUI can be accessed via:
  - http://<ecp-broker-ip>:8161/
- To get further information about Artemis GUI, please refer to the *ECP Installation Guide, chapter Artemis GUI*.

6.4.1.2. Check ecp-broker service status

- Please refer to the chapter *Starting and stopping ECP Broker*.

6.4.1.3. Check Artemis logs

- In case of any issue, it is recommended to check the logs. Logs are located in */opt/ecp-broker/broker/log* folder.
- Application log (Artemis) contains all events related to application behavior such incoming connections and synchronization with Component Directory.
- In case that audit logging is enabled, information such synchronization, service start/stop, certificate expiration and user activities is available there. List of audit log events can be found in the audit log excel, which is part of the ECCo SP Documentation Delivery.

6.4.1.4. Connectivity test to Component Directory

- To test the connectivity between Message Broker and ECP Component Directory check folder for ECP Component Directory content storage. The default folder location is */opt/ecp-broker/cd*. The folder should contain a file **ComponentDirectory.xml** with the last modification date not older than 5 minutes.

6.4.1.5. Artemis console – check of connections from other ECP Endpoints

- Connect to Artemis console on URL *http://<ecp-broker-ip>:8161/*, from the home screen click on the Queues item from the upper menu.
- Check that queues in Artemis console (except DLQ) have a consumer (Consumer Count). Make sure that Message Count is 0 or decreasing.
- If many clients are disconnected, restart ecp-broker service.
- If majority of clients is still disconnected after service restart, validate network rules. Incoming connection to port 5671 needs to be allowed.

- If just several clients are disconnected or not consuming messages from queue, contact owner of ECP Endpoint to restart their ecp-endpoint service and check their logs for a root cause analysis.

6.4.1.6. Certificates expiration check, renewal

- Information about Message Broker certificates expiration is available either in authKeystore.jks file using for instance keytool command:

  keytool -list -v -keystore authKeystore.jks -storepass "password" | grep -m 1 "until"

  or via information available on Endpoint or Component Directory > Components > Detail on Broker component via button [>] > Section Certificates.

- Renew certificates manually if certificates are not expired yet.
  - > To get further information about certificates renewal please refer to the *ECP Administration Guide, chapter Certificates renewal*.
- Send a new registration request to Component Directory if the certificates are already expired.
  - > To get further information about registration to Component Directory, please refer to the *ECP Installation Guide, chapter Registration process*.
  - > Restart of Message Broker is required after the reregistration is done.

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